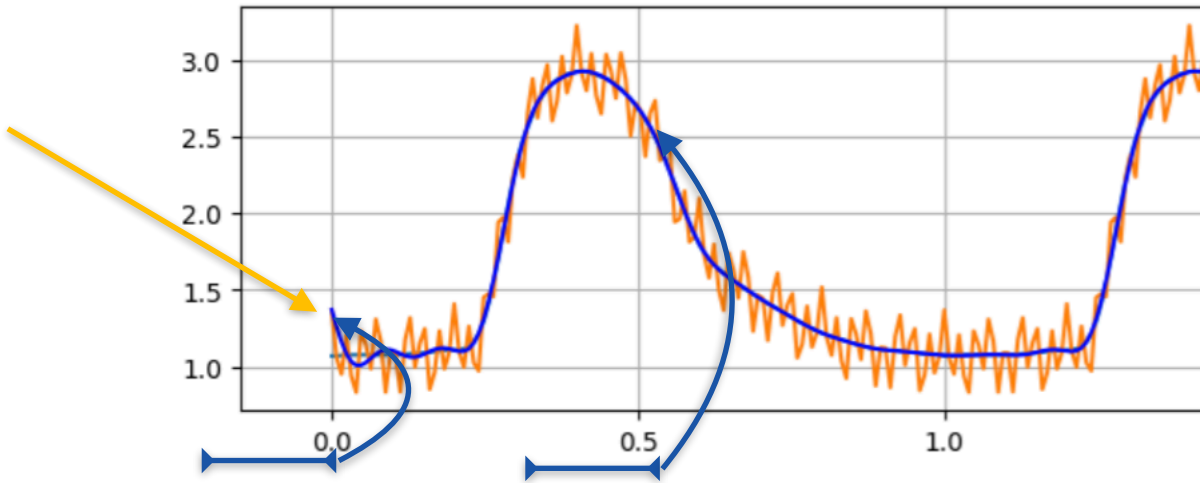


Lecture 4: Measurements, Accuracy, and Statistics

PCC1 (CM2015) - Autumn 2025 - Martin Jacobsson

Why this behaviour?

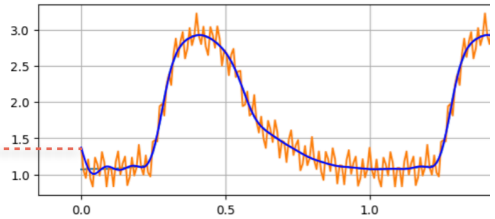


Moving window

Solution: Padding (invent values)

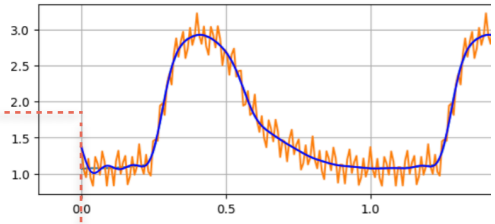
Different types of padding

Extend border value

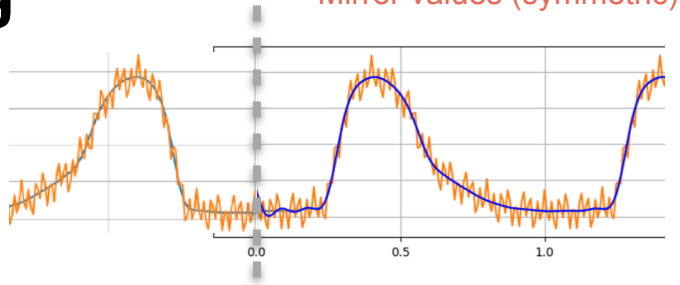


Mean padding

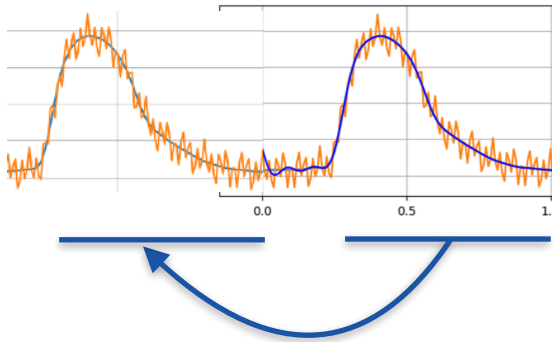
Zero padding



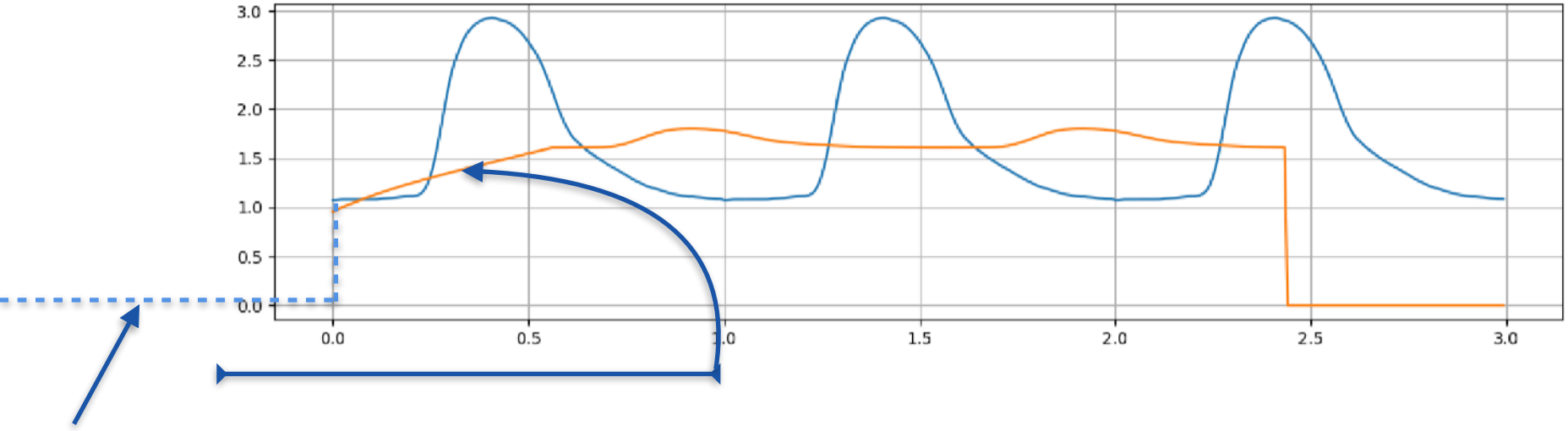
Mirror values (symmetric)



Wrap around values (periodic)
(also assuming sequence is one period)



M=140



Zero padding is used here

Statistics

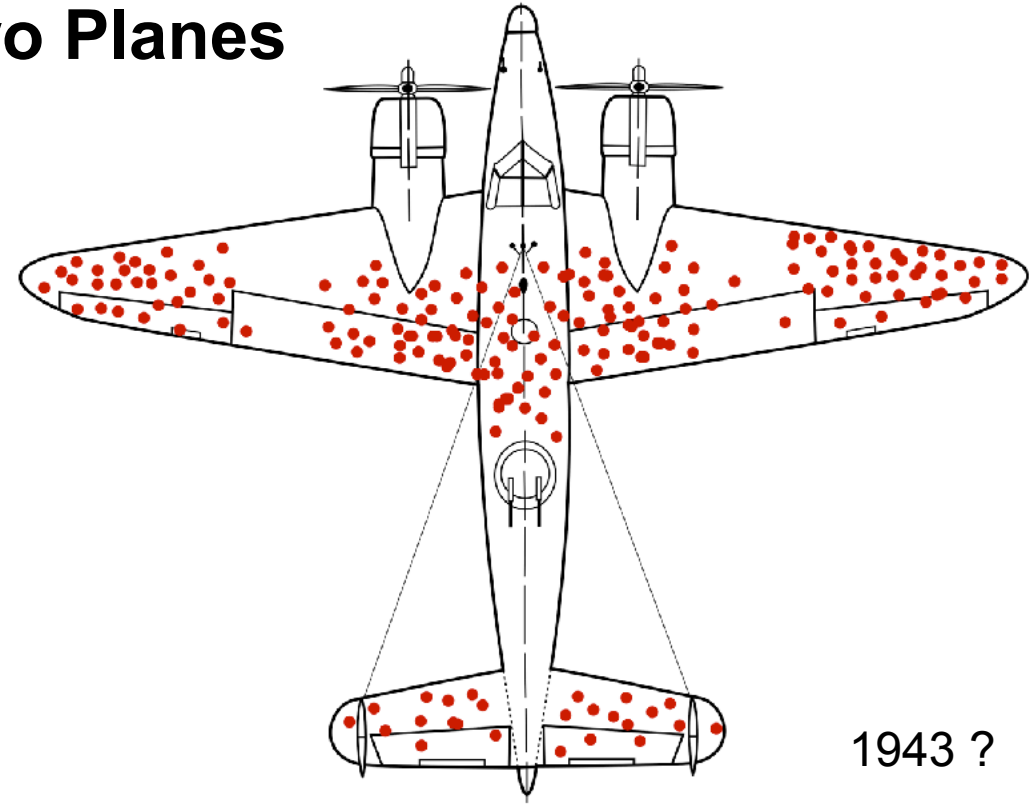
Refreshing

The Tales of Two Planes

Tale 1: Experimental design

Abraham Wald (1902-1950),
Columbia University

Survival bias



1943 ?

Case: Trauma Team Activation

Data from:
Trauma registry

Supervised Machine Learning for Identification of Severely Injured Trauma Patients

Salvar Andri Jóhannsson, Degree project at KTH

Shall we summon the trauma team???



Trauma patient
Situation: …
Background: …
Assessment: …
Recommendations: …
Vital signs values, GCS, …



Also: Oscar Lapidus,

Tale 2: Average is not enough!

US Air Force anecdote

“The average pilot”

4053 pilots, 140 dimensions (weight, length, …)

10 dimensions, 30% tolerance → how many pilots?

Todd Rose, “The End of Average: How We Succeed in a World That Values Sameness”, Harper Collings, New York, January 2016.

“Ability-based design”
https://doi.org/10.1145/3148051

Summary Statistics (1)

Purpose: Reduce all data down to comprehensive information

Central representation

Mean, Median, Mode

Dispersion representation (or scatter, spread)

Variance, Standard Deviation, Min/Max range, quantiles

Summary Statistics (2)

Let **X** be the set of measured values. **N = |X|**

X is called a sample sample ⊆ population

Mean and variance:

- μ and σ^2 for population $\bar{x} \approx \mu$
- $\bar{x}$ and s^2 for sample $s^2 \lessapprox \sigma^2$

Strong law of large numbers $Var(\bar{x}) = \sigma^2/n$

$$P\left(\lim_{n\to\infty} \frac{X_a + \cdots + X_n}{n} = \mu\right) = 1$$

where X_i are i.i.d. (independent and identically distributed)

Central Limit Theorem

How large is a "large number"?

Weak/Strong Law of Large Numbers state that as $n \to \infty$, :

- the mean converges, and
- the variance decreases

$$\bar{X}_n \equiv \frac{X_1 + \cdots + X_n}{n}$$

CLT goes further:

$$\frac{\bar{X}_n - \mu}{\sigma_{\bar{X}_n}} \to \mathrm{N}(0, 1) \quad \text{as } n \to \infty \text{ and } \quad \sigma_{\bar{X}_n} = \frac{\sigma}{\sqrt{n}}$$

CLT states that any arithmetic mean of i.i.d. random variables tends to a normal distribution for large n.

The sum of two normal iid:s is also normal!

Central Representation

Mode — The most frequent value

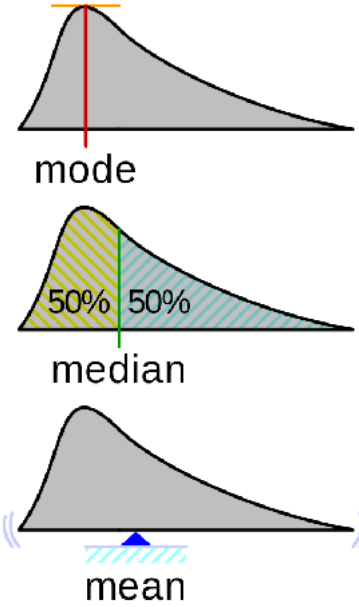
Mean (arithmetic mean, AM)

$$\overline{X} = \frac{\sum_{n=0}^{N-1} X_n}{N}$$

Geometric mean (GM)

$$GM = \sqrt[N]{X_0 X_1 \cdots X_{N-1}}$$

Median — The value “in the middle”



AM-GM Inequality

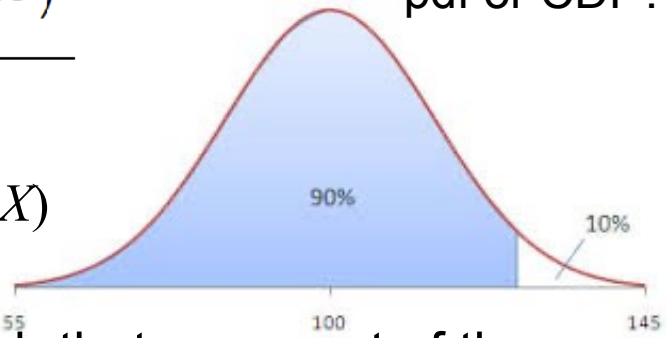
$$\frac{x_1 + x_2 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 \cdot x_2 \cdots x_n}$$

Spread Representation

Variance $\mathrm{Var}[X] = \mathrm{E}[(X - \mathrm{E}[X])^2]$

Standard deviation $s = \sqrt{\frac{\sum_{n=0}^{N-1} \left(X_n - \overline{X}\right)^2}{N-1}}$

pdf or CDF?



Min/Max Range $\mathrm{Max}(X)$ - $\mathrm{Min}(X)$

Percentiles (p) The value such that p percent of the values are less

Other Resources

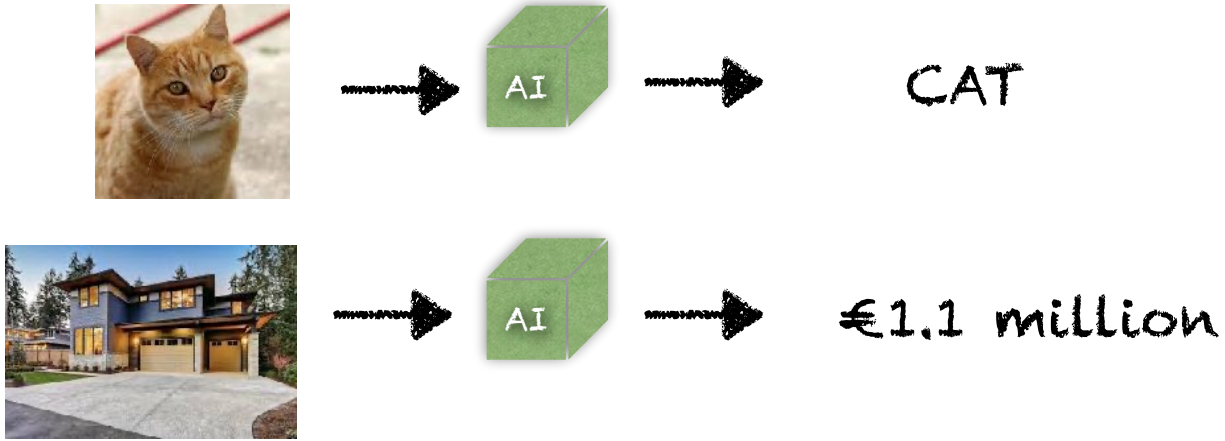
Medical Statistics II: Descriptive Statistics and Standard Scores

https://www.youtube.com/watch?v=9caLf_0bYzo

Accuracy

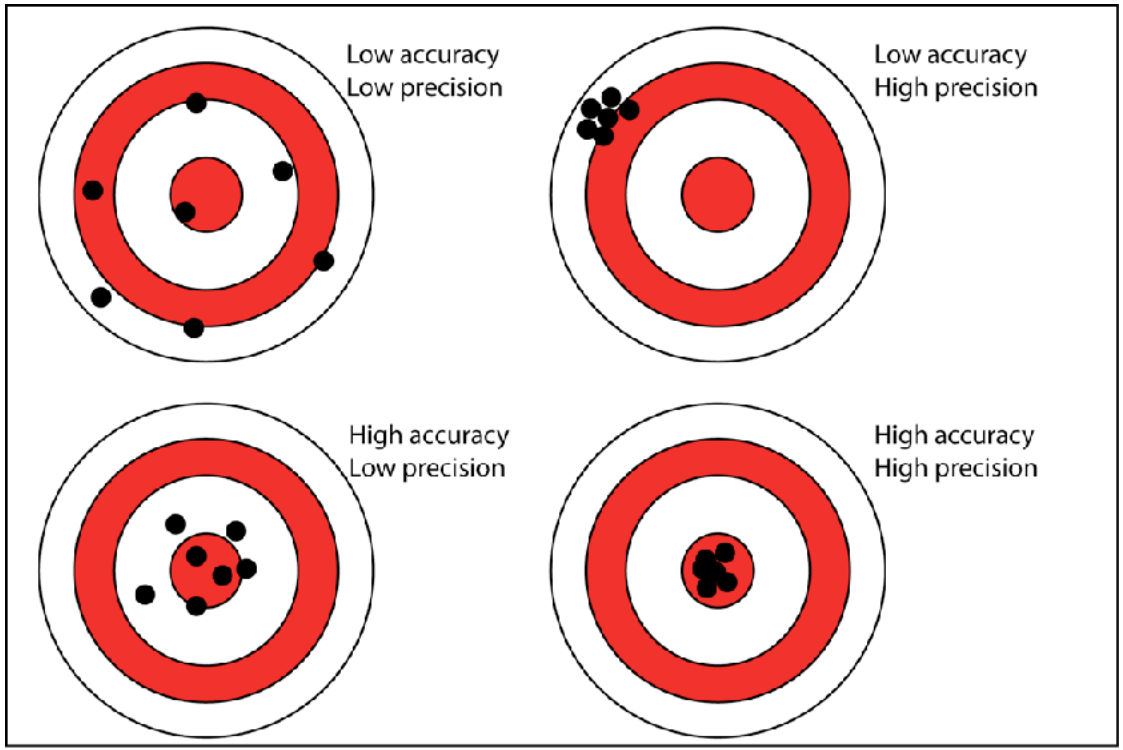
Regression vs. Classification

Linear fit (Ordinary Least-Squares), Logistic Regression, SVM, …



Regression + threshold -> Classification

Precision vs. Accuracy



Measurement Error

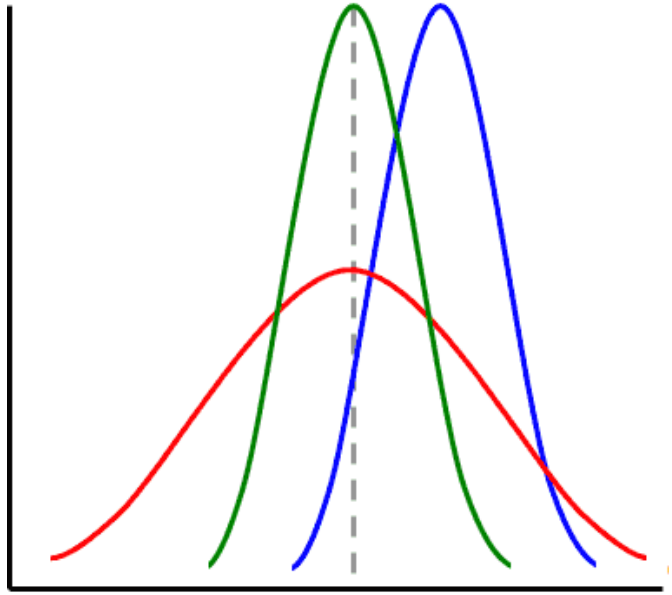
T is the true value
X is the measured value

Usually Gaussian distributed (Normal)

$$X = T + e_s + e_r$$

$e_s \in \boldsymbol{R}$ (constant)

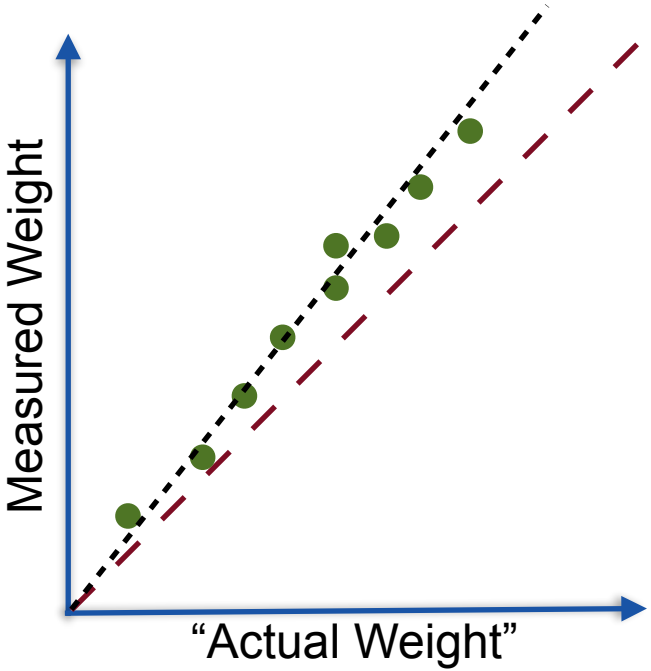
$e_r \in N(0, \sigma)$ (random)



Weak law of large numbers

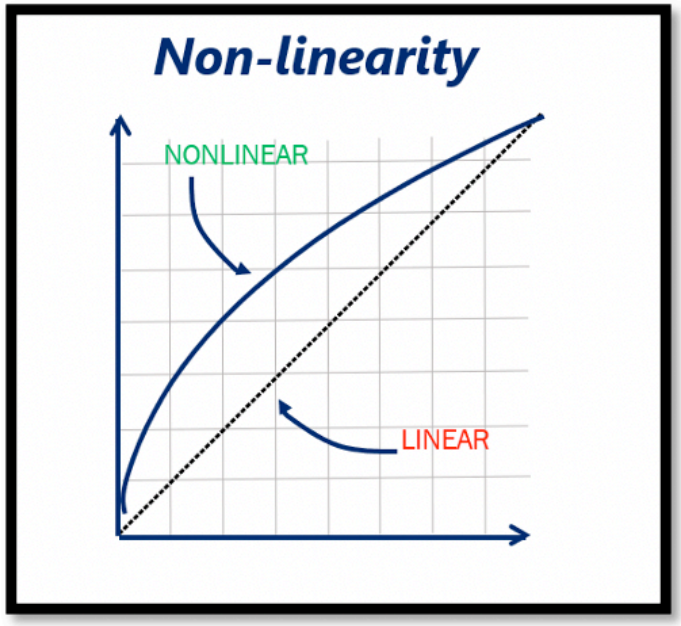
$$\hat{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad \hat{X}_n \xrightarrow{P} T + e_s \quad \text{as } n \to \infty$$

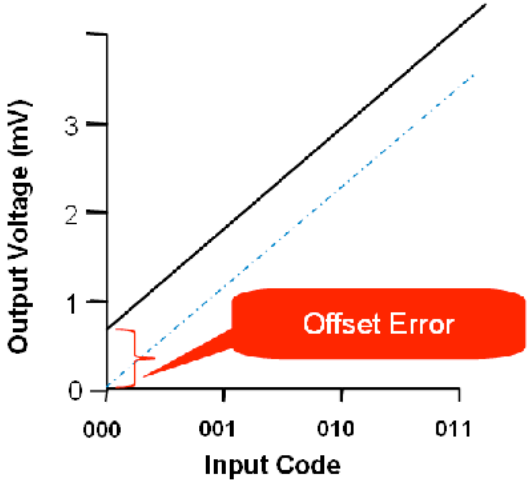
Testing a Scale

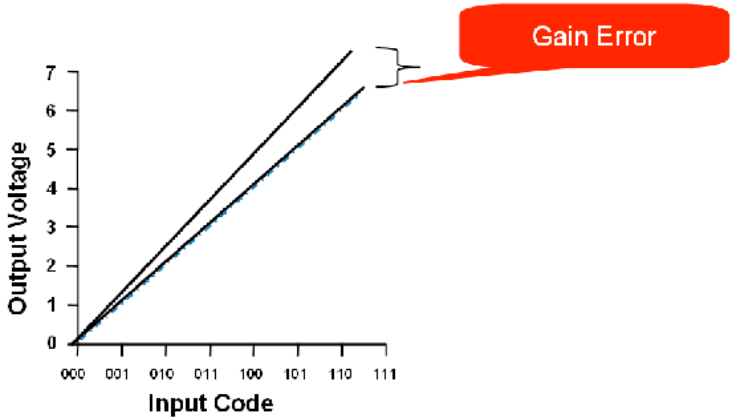




Different types of errors







Measurement Error, More Detailed

$$X = \alpha f(T) + e_s + e(\boldsymbol{Z}) + e_r$$

$e_s \in \boldsymbol{R}$	(systematic error, offset error)
$e(\boldsymbol{Z})$	(error dependent on (un)known factor(s))
$e_r \in N(0, \sigma)$	(residual random error)
$f(x) \approx \mathrm{Id}_X$	(non-linearity error)
$\alpha \approx 1$	(gain error)

Note: σ can be dependent on T.

Misc. Errors

Measurable span (e.g., only 30 - 200 BPM)

Average error vs. Worst-case error

Random errors???

- Reduce variance by repeated measurements?
- Affected by external factors?
- Can those be modelled? Compensated for?
    - Ambient light, Temperature, Moon phase, etc.

Measuring Errors

Mean Squared Errors (MSE)

$$\frac{1}{n}\sum_{i}^{n}(y_i - \hat{y}_i)^2$$

Root MSE (RMSE)

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Mean Absolute Errors (MAE)

$$\frac{1}{n}\sum_{1}^{n}|y_i - \hat{y}_i|$$

Median Absolute Deviation (MAD)

$$\text{median}_i(|y_i - \bar{y}|)$$

Mean Absolute Percentage Error (MAPE)

$$\frac{100}{n}\sum_{i}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

MARD (Mean Absolute Relative Difference)

Term often used in continuous glucose monitoring (CGM)

$$\mathrm{MARD} = \frac{100\%}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

See for instance:

Freckmann G, Mende J, Pleus S, Waldenmaier D, Baumstark A, Jendrike N, Haug C. Mean Absolute Relative Difference of Blood Glucose Monitoring Systems and Relationship to ISO 15197. J Diabetes Sci Technol. 2022 Sep;16(5):1089-1095. doi: 10.1177/19322968211001402. Epub 2021 Mar 24. PMID: 33759584; PMCID: PMC9445334.

Biased estimator

$$\mathrm{E}[X] = \sum_{i=1}^{\infty} x_i \, p_i$$

$$\mathrm{E}[X] = \int_{-\infty}^{\infty} x f(x) \, dx$$

Compare with systematic error, offset error

$$\mathrm{Bias}(\hat{\theta}, \theta) = \mathrm{Bias}_{\theta}[\,\hat{\theta}\,] = \mathrm{E}_{x|\theta}[\,\hat{\theta}\,] - \theta = \mathrm{E}_{x|\theta}[\,\hat{\theta} - \theta\,]$$



- Bias means low accuracy
- Increasing sample size does not improve!

However, we can compensate for the bias! Compare:

Var[Y] ~ $$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2 = \frac{\sum_{i=1}^{n} x_i^2}{n-1} - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{(n-1)n} = \left(\frac{n}{n-1}\right) s_n^2$$

Consistency

An estimator T_n of parameter θ is said to be consistent, if it *converges in probability* to the true value of the parameter:

$$\operatorname{plim}_{n\to\infty} T_n = \theta$$

As the number of data points increases indefinitely, the resulting sequence of estimates converges in probability to θ.

Unbiased, not consistent: for an iid sample $\{x_1, \ldots, x_n\}$ one can use $T_n(X) = x_n$ as the estimator of the mean $\mathrm{E}[X]$.

Biased, but consistent: $T_n(X) = \frac{1}{n}\sum x_i + \frac{1}{n}$ converges to $\mathrm{E}[X]$ as $n\to\infty$

Bias, Consistency, Efficiency

Estimator properties:

- Bias means that we miss the real value, even if sample size is large
- Consistency means that the estimator is better if we increase the sample size
- Efficiency means that it converges faster. Usually measured as MSE (Mean Squared Errors).
- Also robustness (handles outliers in a good way, such as *median* instead of *mean*)

Confidence Interval (CI)

$$CI = \bar{x} \pm z\frac{s}{\sqrt{n}}$$

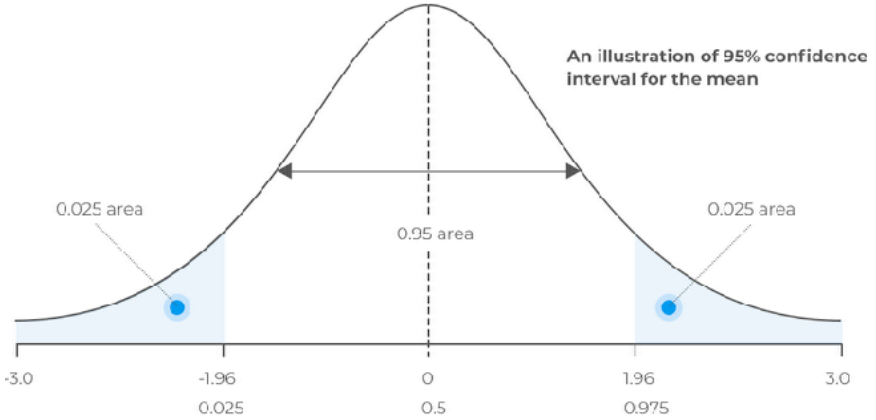
$\bar{x}$ = sample mean

s = sample standard deviation

n = sample size

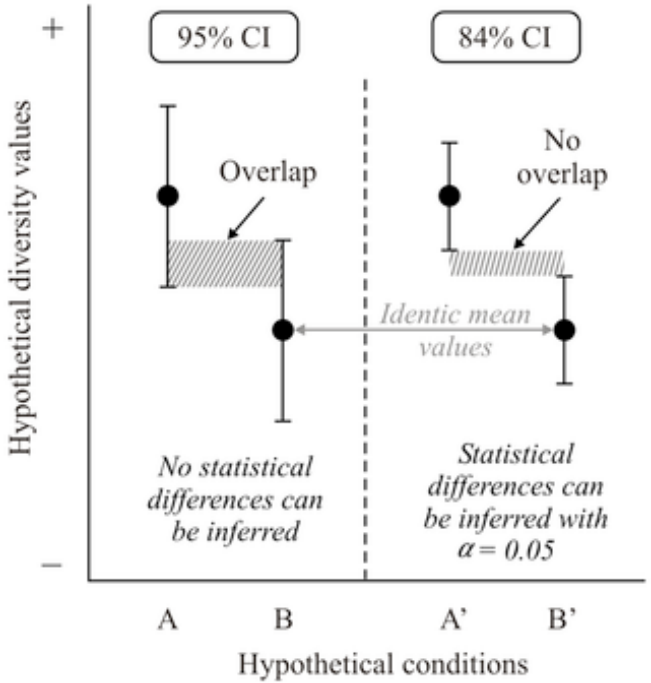
$z \approx 1.96$ assuming 95% confidence and normal distribution

Central Limit Theorem states that summing enough number iid variables tends to normal distribution!
What is enough? Rule of thumb: $n > 20$



Confidence interval is not a dispersion representation!

Overlapping Confidence Intervals



Overlapping? Solutions:

- Lower confidence
- Increase n
- Use hypothesis testing instead

Covariance, Correlation

Pearson's correlation coefficient

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

The values are on or between −1 and 1

Compare with: $\operatorname{cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$

Paired t-test

Paired t-tests have the following hypotheses:

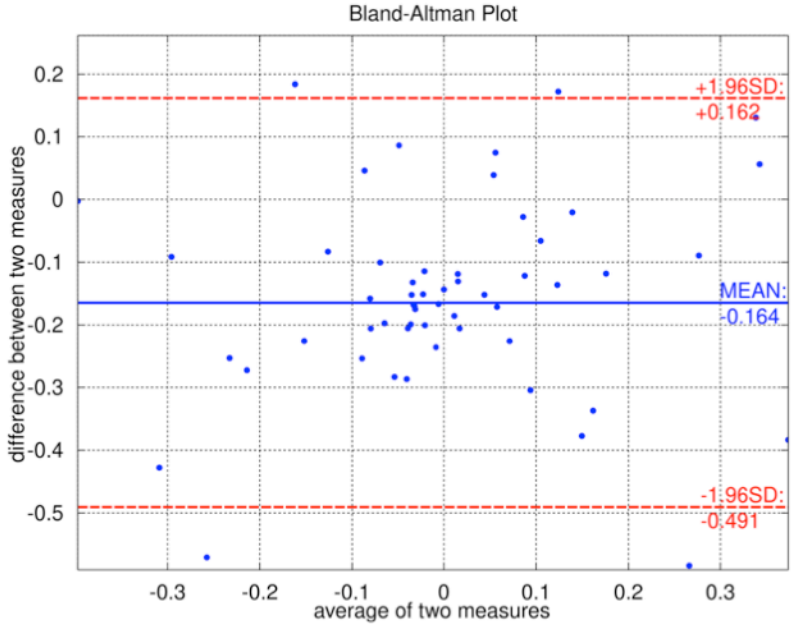
- Null hypothesis: The mean of the paired differences equals zero in the population.
- Alternative hypothesis: The mean of the paired differences does not equal zero in the population.

Assumptions:

- You have a random sample with independent subjects
- Paired sample t-tests use the same items in both groups.
- Your data must be continuous
- Data should follow a normal distribution or sample size ⩾ 20

Bland-Altman (Limits of Agreement)

$LoA = d \pm 1.96 \cdot \sigma_d$ where d is the mean of difference and σ_d is the variance of difference



Intended Use

Intended Use

Important for CE-marking a Medical Device (MDR - Medial Device Regulation - EU 2017/745)

What is the intended use?

- For what purpose?
- On what subject? In what condition/situation?
- How to use the device? By who?



16 990 kr 699 kr

CE-marking

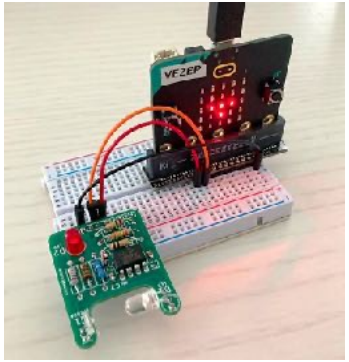
Different classes for different types of medical devices:

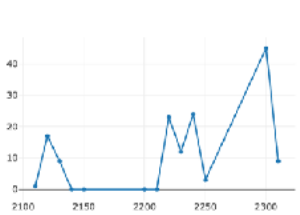
We need to show that device is:

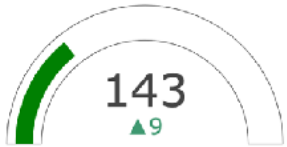
- Safe to use
- Efficient

The latter mean that we need to prove that it can do its job. I.e., it is accurate for all it's intended use.

Lab 7 Tasks







1. Define a crisp clear intended use
2. Define mathematically when the device is “efficient”
3. Design tests to be carried out to prove the above.
4. Carry out at least some of above’s tests.
5. Draw conclusions from this.

It is not necessary that your device passes your rigorous standards!

Exercise

Give examples of applications when this device is not suitable?



Intended Use



Setting (ICU, OR, surgery, sports, …)

Application (HR, HRV, diagnosis of tachycardia, AF, …)

Target user group (adults, children, professionals, …)

Environments (indoor, outdoor cold, sauna, in the pool, bright daylight, darkness, …)

Inclusion (gender, skin color, handicap, amputated, …)

Test Design

If your device includes *children* in its intended use, you need to test on *children*!

If your device includes *sports use* in its intended use, you need to test on *sports use-cases*!

If your device includes *X* in its intended use, you need to test on *X*!

Tests should also mimic the real world application as far as possible! (but still controlled experiment)

Thanks for Today!

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